

KVERNERGET KRISTIANSUND, Norway

WMO: 012230

Lat: 63.1094N

Lon: 7.8158E

Elev: 204

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	21.7	7.8	8.3	23.9	10.7	9.6	25.8	32.7	40.2	29.3	40.9	7.4	90	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	74.8	61.1	70.6	59.3	67.2	57.7	62.9	71.7	60.7	68.0	59.0	64.9	7.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	75.1	65.5	57.4	70.9	63.6	56.0	67.3	61.6	28.4	72.1	27.0	68.3	25.8	64.7	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.9	24.3	20.6	DB	17.2	81.5	4.8	4.1	13.8	84.4	10.9	86.8	8.2	89.1	4.7	92.1	(4)
(5)			WB	15.6	66.2	4.3	2.5	12.5	68.0	10.0	69.5	7.5	70.9	4.4	72.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	34.6	35.6	37.5	42.7	48.6	52.9	57.9	58.0	53.8	46.3	39.6	36.3	(6)	
	DBStd	10.17	5.83	6.01	5.25	6.00	6.53	5.10	5.63	4.54	4.78	5.45	6.36	6.49	(7)	
	HDD50	2525	477	404	388	232	105	22	1	1	16	140	314	425	(8)	
	HDD65	7193	942	824	853	668	509	366	236	224	338	581	762	890	(9)	
	CDD50	833	0	0	0	15	61	108	246	248	129	24	2	0	(10)	
	CDD65	26	0	0	0	0	1	2	16	7	0	0	0	0	(11)	
	CDH74	143	0	0	0	0	6	11	105	21	0	0	0	0	(12)	
	CDH80	15	0	0	0	0	0	0	15	0	0	0	0	0	(13)	
	WSAvg	8.7	9.2	8.9	10.3	8.2	8.4	9.2	8.1	7.3	8.5	8.4	8.7	9.5	(14)	
	PrecAvg	64.4	6.8	5.9	5.7	3.5	3.1	4.1	4.2	5.4	6.0	6.0	6.0	7.7	(15)	
(16) Precipitation	PrecMax	77.4	16.6	11.7	12.4	10.3	5.8	7.2	7.0	10.8	15.4	11.0	15.0	15.0	(16)	
	PrecMin	43.5	0.2	0.4	1.9	0.9	1.3	1.4	1.7	1.7	2.2	1.3	0.4	2.0	(17)	
	PrecStd	8.1	3.7	2.8	2.8	2.1	1.1	1.6	1.4	2.1	3.2	2.2	3.9	3.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	50.8	52.3	67.2	74.1	75.4	82.0	77.0	69.5	63.9	55.5	52.6	(19)	
		MCWB	42.3	43.4	42.2	52.1	56.9	60.3	66.0	61.8	58.5	52.6	49.1	46.2	(20)	
	2%	DB	46.6	47.6	49.0	60.6	67.8	69.7	77.0	72.6	66.1	60.4	52.7	49.1	(21)	
		MCWB	41.1	41.2	41.5	48.4	54.9	57.6	63.8	59.8	55.8	50.6	46.5	43.5	(22)	
	5%	DB	44.7	45.6	47.0	55.3	63.0	64.7	71.6	68.4	63.0	56.0	50.1	47.0	(23)	
		MCWB	40.5	40.0	41.1	46.3	52.9	55.4	61.3	58.6	55.1	48.4	44.8	42.2	(24)	
	10%	DB	42.7	43.7	45.0	51.4	59.0	61.0	66.9	65.5	60.7	53.6	48.0	44.8	(25)	
		MCWB	38.9	38.8	40.1	43.6	50.3	54.0	59.3	58.1	54.1	47.6	43.3	40.2	(26)	
	0.4%	WB	45.0	45.6	46.0	52.8	58.8	61.1	67.4	63.5	60.3	54.4	50.5	48.1	(27)	
		MCDB	47.0	48.7	47.8	66.2	70.5	73.0	79.6	72.3	66.0	59.0	53.9	50.6	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.2	43.0	44.4	49.3	55.7	58.2	64.5	61.4	58.1	51.9	48.1	45.4	(29)	
		MCDB	45.5	45.4	46.8	58.8	65.7	67.1	75.4	68.8	62.6	57.4	51.4	47.7	(30)	
	5%	WB	40.9	41.3	42.3	46.9	53.5	56.2	62.0	59.9	56.4	50.1	45.8	42.8	(31)	
		MCDB	43.2	44.1	45.8	53.2	62.2	63.3	70.0	66.0	60.8	55.6	48.8	46.0	(32)	
	10%	WB	39.0	39.4	40.5	44.9	51.3	54.5	59.9	58.5	55.0	48.2	43.8	40.4	(33)	
		MCDB	42.3	42.9	44.0	49.3	58.1	60.3	66.0	63.8	59.5	52.3	47.1	43.6	(34)	
	5% DB	MDBR	6.8	7.8	9.5	11.0	11.9	10.0	10.5	10.5	9.9	8.9	7.3	7.2	(35)	
		MCDBR	9.1	9.8	11.5	17.0	18.8	16.5	18.9	16.1	12.7	11.6	9.5	10.2	(36)	
		MCWBR	6.7	6.4	7.5	8.5	9.1	8.0	8.5	7.7	6.8	7.0	6.5	7.9	(37)	
		MCWBR	8.3	8.5	9.9	15.0	17.9	15.0	17.6	13.2	11.5	10.9	9.2	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.242	0.300	0.327	0.349	0.348	0.350	0.358	0.349	0.332	0.308	0.375	0.357	(40)	
		taud	2.073	2.266	2.333	2.388	2.457	2.487	2.487	2.512	2.541	2.451	2.708	2.567	(41)	
	Ebn at Noon		121	203	245	265	276	277	271	264	246	203	118	66	(42)	
		Edh at Noon	10	19	26	30	30	30	30	26	21	16	9	6	(43)	
(44) All-Sky Solar Radiation	RadAvg		46	200	575	1112	1512	1596	1470	1130	670	296	74	24	(44)	
	RadStd		8	36	85	102	78	153	129	106	81	43	16	2	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (32 neighbors)	+0.77	+1.21	N/A	N/A	N/A	N/A	-250	-329	+104	N/A	

Nomenclature: See separate page